



جامعة الكفيل
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Oral Surgery Third stage

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Oral Surgery

Techniques of Dental Local Anesthesia

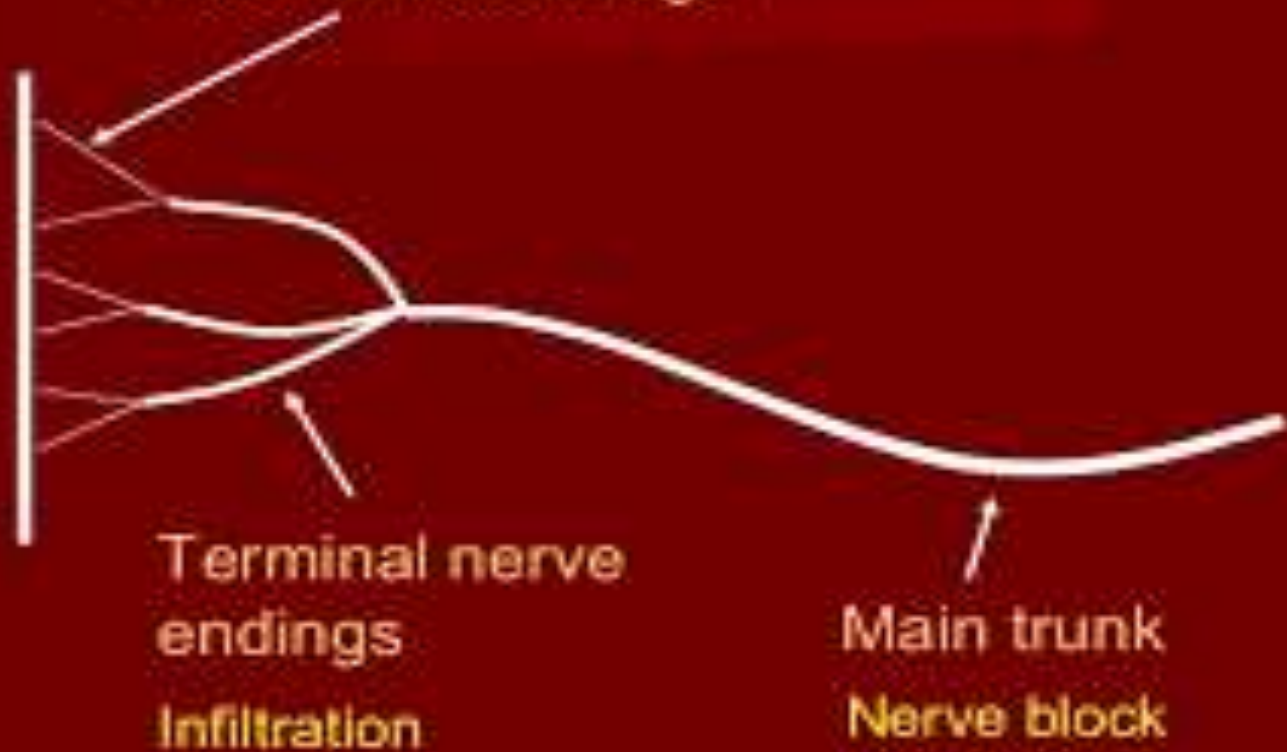
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Techniques of dental Local Anesthesia

- Topical or surface anesthesia technique.
- Infiltration technique.
- Block technique.

Topical
Free nerve endings



Topical or surface anesthesia technique

- It is most commonly used to :
- Obtain anesthesia of mucosa prior to L.A. injection to minimize the pain during needle insertion.
- Prior to carrying out incision & drainage of abscesses.
- Prior to sutures removal.
- Prior to taking full mouth impression for edentulous.

Infiltration anesthesia

- *Maxilla*

Infiltration technique is very effective in maxilla, since the maxilla has thin labial \ buccal cortical plate , shows area of porosity and the compact bone presents numerous foramina which aid in absorption of local anesthetic solution.

- ***Mandible***
- The bone is generally dense and has thicker cortical plates than maxilla, particularly in posterior region.
- Only the anterior part of mandible presents sufficient porosity, which is favorable for infiltration techniques.

- **The advantages of infiltration techniques are:**
- Easy and simple injection
- Very high success rate.
- Good control of bleeding.
- Low risk of complications.

Infiltration technique is subdivided into:

- 1.Sub-mucous injection:**
- 2.Supra-periosteal injection:**
- 3.Sub-periosteal injection:**
- 4.Intra-osseous injection:**
- 5. Supplementary injections**

Peridental injection

Injection in
interdental papilla

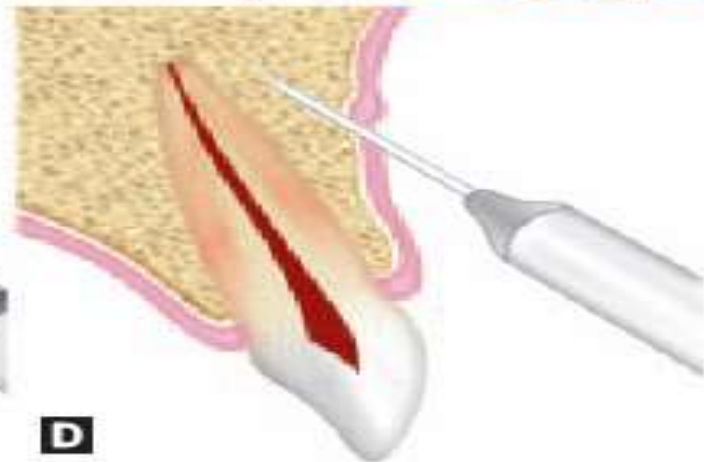
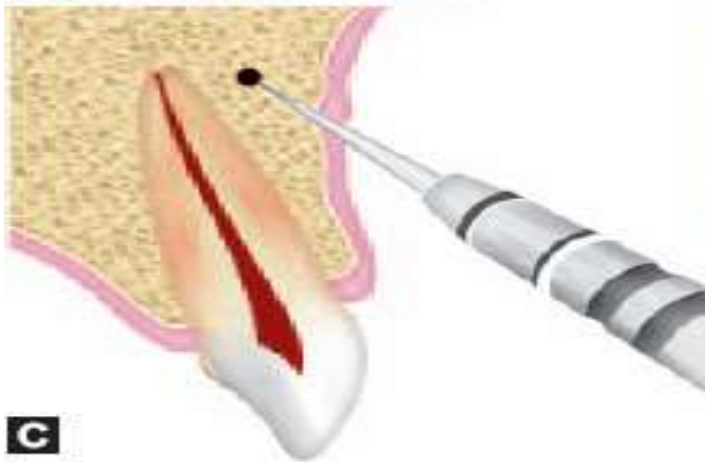
Sub-periosteal
injection

Supra-periosteal
injection

Submucous
injection

A

B



1.Intra-ligamentary:

- In this technique, small amount of anesthesia is placed in the gingival sulcus into the periodontal ligament using small amount of L.A. solution (0.2 ml), then the solution is carried from the alveolar bone and through the apical foramen into the pulp chamber.

1. Intra-ligamentary:

- This technique is most frequently used in:
- 1. Patients still feel pain despite the normal techniques of infiltration or block.
- 2. when bilateral treatment is planned.
- 3. pediatric or disabled patients when there is concern of postoperative tissue trauma to the lip or tongue.

Disadvantage

- Discomfort or pain
- Tissue damage or irritation
- Incomplete Anesthesia



2-Intrapulpal injection:

- . This technique is used to control pain during procedures which required direct instrumentation of the pulpal tissues and when anesthesia from other techniques is inadequate.
- The needle should be hold firmly into the pulp chamber or root canal, slight discomfort is felt which subsequently gets subsided.



Infiltration technique of the maxilla :

Anesthesia of the upper anterior teeth

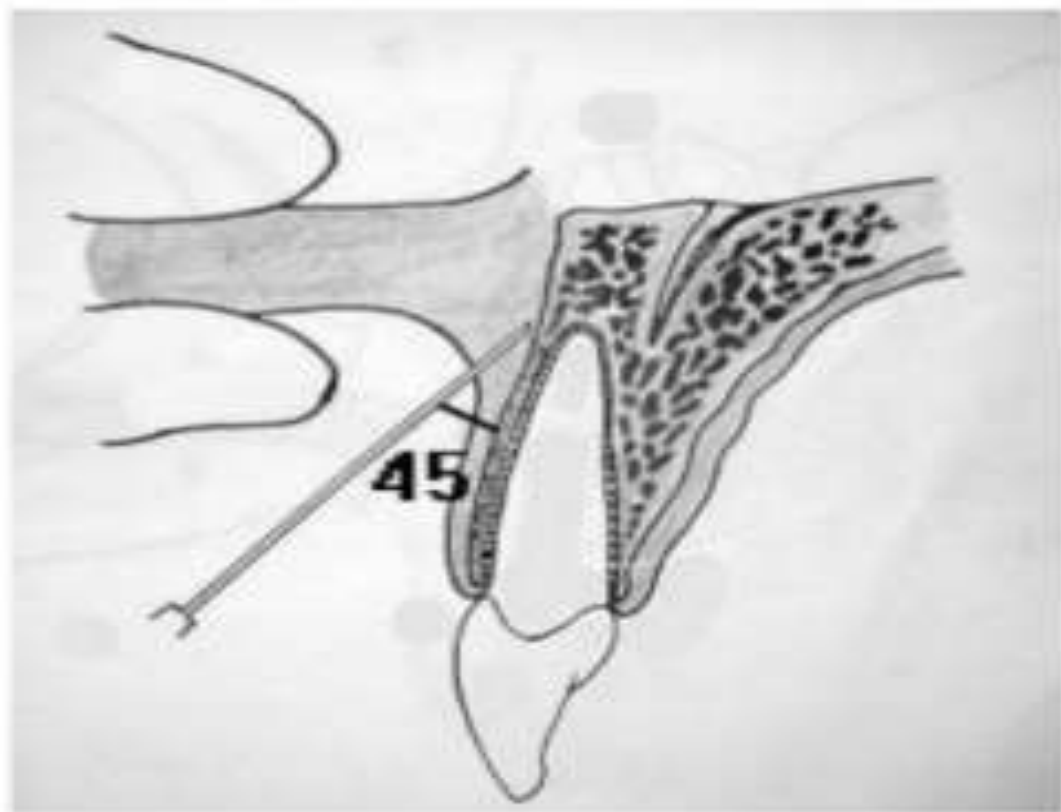
- The central, lateral incisor, and canine teeth and the labial supporting tissues and muco-periosteum related to them are innervated by the anterior superior alveolar nerve (supplies the pulp and soft tissue labially). These teeth are anaesthetized by infiltration injection by depositing about **1ml.** of local anesthetic solution near the apex of tooth concerned as follows.

Buccal mucosa Parotid papilla Mucobuccal fold Buccal frenum



- 1. Hold the syringe parallel with the long axis of the tooth.
- 2. Insert the needle into the height of the mucobuccal fold over the target tooth at an angle of 45 degree to the long axis of the tooth.
- 3. Advance the needle few millimeters.
- 4. inject slowly.
- 5. slowly withdraw the syringe.





Anesthesia of the palatal tissues

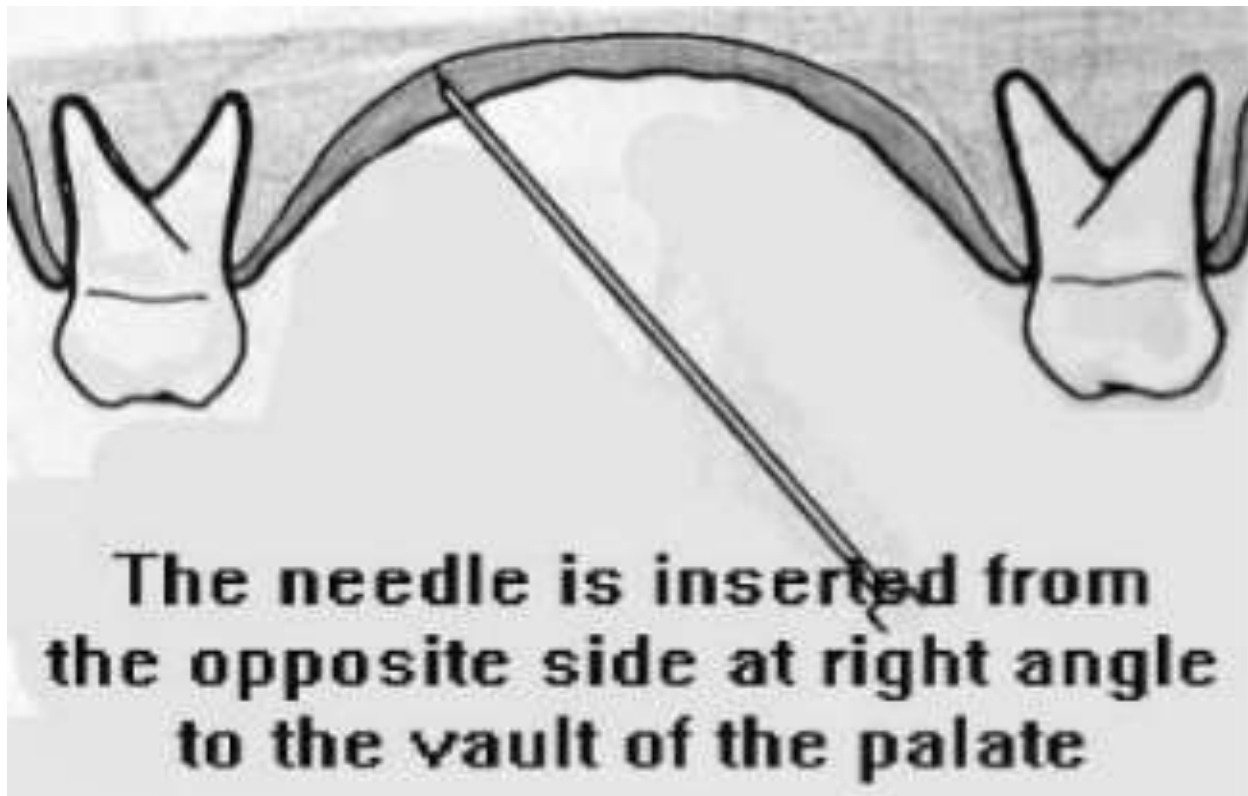
- The palatal muco-periosteum is firm and is closely adapted to the underlying bone. These characteristics make it necessary to inject local anesthetic solution under greater pressure. For this reason patients should be informed prior to injection that palatal injections cause some discomfort or pain.

Anesthesia of the palatal tissues

- This can be minimized by inserting the needle with the bevel facing the bone and as near as possible at right angles to the vault of the palate 90°. The solution is deposited in the palatal tissue about half way between the midline and the gingival margin of the target tooth.

- When the needle touches the bone, gently, withdraw the needle by 1 mm (to avoid painful subperiosteal injection) and deposit about 0.25 – 0.5 ml of the solution in the area to be anesthetized, withdraw the needle slowly.





Anesthesia of the premolar teeth

- The mesio-buccal root of the 1st molar, both premolars, and the buccal supporting tissues and muco-periosteum related to them are innervated via the middle superior alveolar nerve (supplies pulp and soft tissue buccally).

Anesthesia of the premolar teeth

- Infiltration techniques are used to anesthetize these structures by deposition of 1 ml of solution near the apex of tooth concerned.

We should also give anesthesia to the greater palatine nerve which supplies the palatal mucosa of these teeth in the same way mentioned previously.

Anesthesia of molar teeth

- The 3rd molar, 2nd molar, the disto-buccal & palatal roots of the 1st molar and their buccal supporting tissues and muco-periosteum are innervated by the posterior superior alveolar nerve (supplies pulp and soft tissue buccally).

Anesthesia of molar teeth

- Infiltration techniques are used to anesthetize these structures by deposition of 1 ml of solution near the apex of tooth concerned. We should also give anesthesia to the greater palatine nerve which supplies the palatal mucosa of these teeth in the same way mentioned previously.

Block technique of the maxilla :

- Since infiltration injections are so effective in the maxilla, block injections are rarely used, but may be of value if:

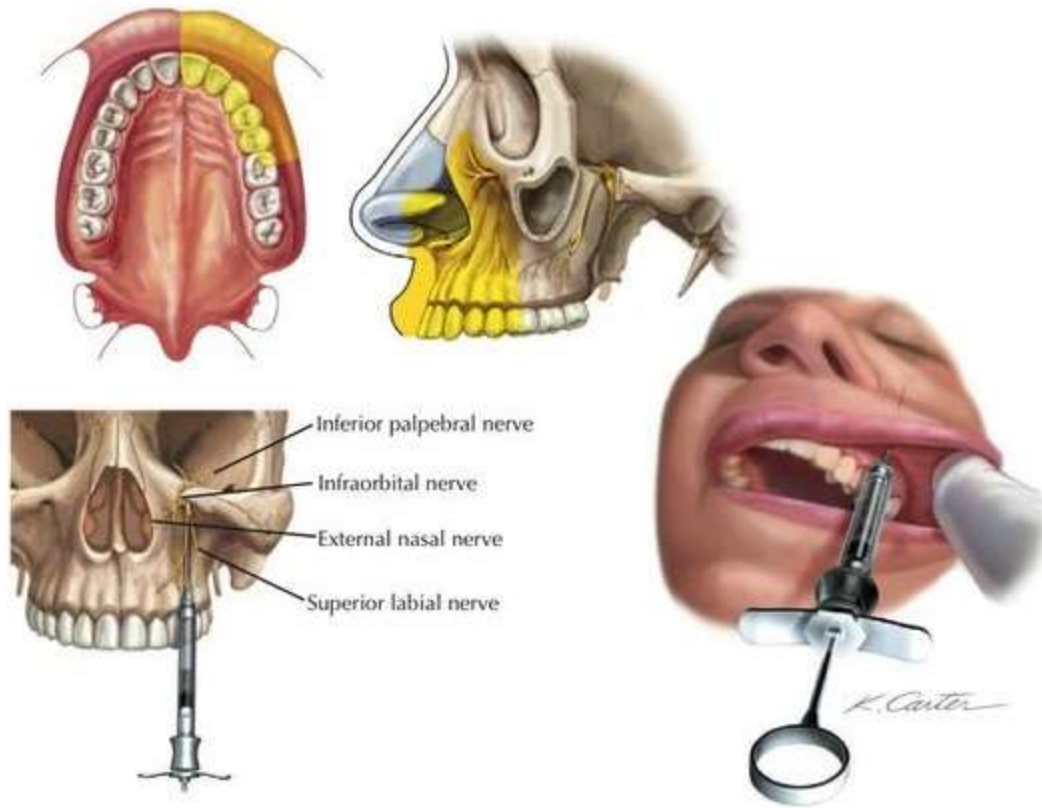
- Extensive or multiple extraction is to be done.
- Surgical procedures.
- The presence of infection at the site of injection where infiltration cannot be used.

The infra-orbital block injection:

- Technique:
- * The infraorbital ridge is palpated and the infraorbital notch is located with the tip of the 1st finger.
- * then moving it slightly downwards to lie directly over the infraorbital foramen.
- with the fingertip maintained in its position the thumb reflect the upper lip & expose the site of injection.

The infra-orbital block injection:

- * needle is inserted into the height of the mucobuccal fold over the apex of the 2nd premolar parallel to its long axis
- * The needle is advanced in line with the long axis of the 2nd premolar to a depth of 1.5-2 cm , 1 ml is deposited.
- * the correct position is confirmed when the injected solution is felt beneath the fingertip.



The posterior superior alveolar nerve block

- In this technique the solution deposited close to the posterior superior alveolar nerve after it leave its bony canal thus producing anesthesia of the 3rd molar, 2nd molar and the 1st molar except the mesio-buccal root and their supporting structures.

Technique:

- Partial opening of the mouth
- Retracting the cheek with your finger
- Inserting the needle into the height of the muco-buccal fold over the 2nd molar
- Advance the needle in the upward, inward and backward direction in one movement.
- 1 – 1.8 ml of solution is deposited.

Disadvantage

- Possibility of damaging the pterygoid venous plexus and the risk of hematoma formation.



The greater palatine nerve block:

Technique:

Needle is injected slightly anterior to the greater palatine foramen and care should be taken to avoid injecting too close to the greater palatine foramen as the lesser palatine nerve may be affected which produce anesthesia to the soft palate, tonsillar and uvular areas making swallowing difficult and causing discomfort or distress to the patient, so that should never be given injection posterior to the 2nd molar.

- The needle is inserted in the palatal soft tissue distal to the maxillary 2nd molar about 1cm from the palatal gingival margin toward the midline, deposit 0.25 to 0.5 ml of L.A. solution very slowly.



- **The nasopalatine nerve block:**
- This technique provide anesthesia to the anterior portion of the hard palate (soft & hard tissues) from the mesial of the right 1st premolar to the mesial of the left 1st premolar.
- Technique:
- Needle is inserted to the palatal mucosa just lateral to the incisive papilla.



Thank
you